

Small-Data Surrogate Shock-Aware Designed (S3) models: Gradient-Free Residual Adaptation with Adaptive Conformal Uncertainty for Engineering Time Series

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Abstract

Forecasting short engineering time series is difficult because limited observations, non-stationary dynamics, and calibration drift make high-capacity local models unstable and can also reduce the reliability of zero-shot priors. This study introduces Small-Data Surrogate shock-aware design (S3), a gradient-free residual-adaptation framework comprising two complementary architectures. S3-Forecaster combines a frozen causal prior with a fixed Echo State Network, a Bayesian Ridge residual readout, and a volatility-sensitive out-of-bag gate. S3-FastSketch replaces the recurrent reservoir with deterministic causal multiscale features, a closed-form Ridge readout, and bounded calibration shrinkage. Both variants are coupled with Adaptive Conformal Inference for sequential prediction intervals. The methods are evaluated under a strict chronological train–calibration–test protocol on an anonymized industrial construction-material demand series and four public monthly benchmarks, with all histories containing fewer than 200 observations. Against statistical, kernel, deep-learning, lightweight, spline-based, and foundation-model baselines, the results show dataset- and objective-dependent advantages rather than universal dominance. S3-Forecaster achieves the lowest root mean squared error and highest coefficient of determination on the industrial application, whereas S3-FastSketch attains the lowest proportional error. On the M4 benchmark, S3-FastSketch is the strongest point forecaster, while S3-Forecaster yields the best adaptive conformal interval score. Ablation results show that residual adaptation is beneficial only when prior errors contain stable predictable structure, and that gating mainly acts as protective shrinkage against over-correction. The proposed family therefore provides a computationally efficient adaptation layer for short, non-stationary engineering series without end-to-end neural retraining.

Keywords: small-data forecasting, residual adaptation, reservoir computing, conformal prediction, non-stationarity, industrial forecasting

1. Introduction

Time-series forecasting is a control-relevant inference layer in complex engineering systems because its outputs are consumed directly by production planning, automated inventory allocation, and logistics optimization modules rather than being used only as descriptive summaries [1, 2]. The same dependence appears in smart-grid operation and predictive maintenance, where forecast errors propagate into dispatch imbalance, downtime, reliability losses, and suboptimal asset scheduling [3, 4]. In modern industrial pipelines,

sensor streams, SCADA records, enterprise transactions, and maintenance logs are fused into rolling forecasting services whose predictions feed downstream optimization and control loops across multiple horizons [5]. The immediate collision with reality is that many asset-level or SKU-level engineering series are not data-rich: new installations, catalog churn, sparse failures, intermittent demand, and local operating reconfigurations often leave only short and irregular histories from which the forecaster must generalize [6].

Formally, let $\{y_t\}_{t=1}^N$ denote a univariate engineering series and let $h \geq 1$ be the forecast horizon. In the small-data regime considered here, $N < 200$, and the effective number of supervised lag–target tuples after windowing with lag length w is only

$$M = N - w - h + 1,$$

which is frequently too small to identify a high-capacity predictor f_θ without severe estimation variance [5, 7]. Representative state-of-the-art families range from recurrent LSTMs and efficient attention models such as Informer to patch-based Transformers such as PatchTST [8–10]. At the foundation-model end, Chronos and TimesFM replace local estimation by massive global pretraining and zero-shot transfer across heterogeneous collections of time series [11, 12]. However, when one isolated industrial series is used as a stand-alone training task, end-to-end estimation becomes statistically fragile because the parameter-to-sample ratio is unfavorable, regularization becomes underspecified, and local validation itself is noisy under such short histories [7]. Benchmark evidence has consequently shown that architectural sophistication is not a sufficient condition for better forecasting, since simple linear baselines can outperform many Transformer forecasters, while non-stationary settings have been reported to induce severe overfitting in attention-based designs [13, 14]. For globally pretrained models, the prior is powerful only when source–target similarity is adequate; otherwise the imported weights can miss the microscopic local dependencies of a single engineering series and induce bias, uncertainty distortion, or outright negative transfer rather than adaptation [15].

A second bottleneck is the coexistence of non-stationary dynamics and structural shocks, which is more faithfully represented by

$$y_t = m_t + a_t + \sigma_t \varepsilon_t, \quad \varepsilon_t \sim \mathcal{D}_t(0, 1),$$

where the latent level m_t , local amplitude a_t , conditional scale σ_t , and even the innovation law $\mathcal{D}_t$ may jump at unobserved break times because of maintenance interventions, control-policy modifications, promotions, supply disruptions, or exogenous demand shocks [16, 17]. Traditional remedies address this instability with smoothly evolving latent states or recursively discounted parameters, including time-varying parameter regressions, parsimonious TVP-VAR constructions, and related dynamic Bayesian formulations that are effective under gradual drift but structurally misaligned with abrupt nonlinear transitions [18, 19]. A complementary line relies on dynamic model averaging, regime-switching filters, or Bayesian online changepoint detection to detect breaks and then reset or reweight predictors once sufficient post-shock evidence has accumulated [20, 21]. Yet these mechanisms are evidentiary rather than instantaneous: if a break occurs at time τ and a width- w adaptive estimator has seen only k post-shock observations, then a stale mass $1 - k/w$ of the estimator still comes from the obsolete regime, so the fitted parameters remain convex mixtures of incompatible dynamics

and the forecast incurs a systematic geometric lag [23, 24]. Recent online-forecasting studies now identify precisely this adaptation lag as a central failure mode, since rolling updates and online ensembles can exhibit continuous performance regression whenever concept drift accumulates faster than the model can safely relearn [25, 26]. The difficulty is sharper in forecasting than in ordinary online regression because labels arrive only after the horizon closes, so the update rule may adapt to stale concepts rather than to the active operating regime, a temporal mismatch that has motivated proactive drift-compensation mechanisms [27]. In narrow engineering decision windows, this delay produces transient mean-squared-error amplification exactly where forecast instability is hardest to diagnose and most operationally expensive [22, 28].

A third bottleneck is uncertainty quantification. In automated industrial decision-making, a point forecast $\hat{y}_{t+h|t}$ is rarely sufficient unless it is accompanied by a predictive interval or set

$$C_{t,h}^{1-\alpha} = [\hat{y}_{t+h|t} - q_t^\alpha, \hat{y}_{t+h|t} + q_t^\alpha],$$

whose coverage is reliable under real deployment shift rather than only under retrospective fitting [29]. Classical parametric UQ remains useful but brittle in this operating regime: standard Gaussian-process regression assumes homoscedastic Gaussian residuals unless explicitly extended, while baseline GARCH-style volatility recursions impose a low-dimensional parametric evolution on conditional variance and do not by themselves resolve skewness, heavy tails, or break-driven distributional change [30, 31]. Indeed, dedicated GP extensions are required for heteroscedastic non-stationarity, and empirical studies show that skewness and heavy tails matter materially for predictive performance in stressed regimes relative to Gaussian structural shocks [32, 33]. Conformal prediction is attractive because it replaces a full parametric residual law by calibration through nonconformity scores and thereby targets distribution-free predictive validity under minimal assumptions [34]. However, the basic conformal premise of exchangeability is systematically violated in sequential engineering data with deep temporal dependence and evolving regimes, so vanilla conformal calibration cannot be transferred unchanged to this setting [35, 36]. Adaptive Conformal Inference (ACI) addresses this exchangeability violation by recasting miscoverage control as an online learning problem in which the calibration level is updated sequentially from recent forecast errors rather than fixed once and for all [37, 38]. In practical deployment, such sequential calibration must remain strictly chronological; otherwise future residuals leak into the calibration buffer, creating information leakage and look-ahead bias in both interval assessment and model selection [39, 40].

Motivated by these three constraints, we introduce a unified forecasting family consisting of *S3-Forecaster* and *S3-FastSketch*; throughout this article, *S3* denotes *Small-Data Surrogate shock-aware design*. The design is intentionally asymmetric and can be written as

$$\hat{y}_{t|t-1}^{S3} = \hat{y}_{t|t-1}^{\text{base}} + g_{t-1} \hat{\Delta}_{t|t-1}, \quad 0 \leq g_{t-1} \leq 1,$$

where a causal base predictor is frozen, $\hat{\Delta}_{t|t-1}$ is a gradient-free residual surrogate, and g_{t-1} is a shock-aware contraction factor. First, this directly attacks the small-data bottleneck because S3-Forecaster computes the residual correction from a fixed-reservoir Echo State Network whose fading-memory dynamics are regulated

by spectral radius constraints, while leaving only the readout trainable [41, 42]. S3-FastSketch follows the same low-variance philosophy without recurrence, replacing the reservoir by deterministic multiscale causal features and a convex linear decoder, a design aligned with the growing evidence that carefully constrained linear or decomposition-aware correctors can remain highly competitive when local data are scarce [7, 13]. In both variants, the trainable stage is reduced to a convex readout, so the corrective mechanism inherits convex optimization stability rather than the nonconvex fragility of backpropagation through a deep sequence model [43]. Second, both variants are made shock-aware through out-of-bag calibration gates or bounded shrinkage factors that suppress residual over-correction immediately after abrupt transitions, thereby targeting adaptation lag without repeatedly refitting a large network after each shift [26, 27]. Third, uncertainty is handled by a sequential ACI layer that updates interval width from recent nonconformity scores and seeks coverage robustness under drift without assuming Gaussianity, homoscedasticity, or exchangeability [36, 37]. By separating training, calibration, and deployment chronologically, the proposed S3 family is constructed to minimize information leakage, eliminate look-ahead bias, and reduce negative transfer from obsolete pre-shock segments while preserving an operationally light forecasting workflow [39, 40].

1.1. Key Contributions

1. We formulate small-data engineering forecasting as a *surrogate residual adaptation* problem and propose a gradient-free correction architecture that freezes the baseline forecaster and learns only a low-capacity residual readout, thereby prioritizing variance control and convex optimization stability over end-to-end parameter proliferation [7, 13].
2. We introduce two complementary model realizations—*S3-Forecaster*, based on fixed-reservoir Echo State dynamics under spectral radius constraints, and *S3-FastSketch*, based on deterministic causal multiscale sketches—to separate temporal representation from lightweight residual decoding in genuinely short-series regimes [41, 42, 44].
3. We develop a shock-aware residual control mechanism based on out-of-bag gating and bounded shrinkage, motivated by the documented adaptation lag of classical online updaters under concept drift and by the need to suppress variance amplification when apparent residual structure is not stable enough to be learnable [25, 26].
4. We integrate a sequential Adaptive Conformal Inference layer for uncertainty quantification, so that prediction intervals are calibrated by recent nonconformity scores and retain distribution-free predictive validity despite exchangeability violation in dependent engineering sequences [29, 36, 37].
5. We adopt a strict chronological train–calibration–test protocol designed to preserve temporal order and avoid information leakage and look-ahead bias, aligning evaluation with the actual decision task faced in forecast deployment rather than with i.i.d. cross-validation surrogates [39, 40].

2. Materials and Methods

2.1. Problem Formulation

Let

$$\mathbf{y}_{1:N} = (y_1, \dots, y_N)^\top \in \mathbb{R}^N \quad (1)$$

denote a univariate engineering time series with monthly sampling frequency and limited historical availability, $N < 200$. The forecasting problem is defined with respect to the natural filtration

$$\mathcal{F}_t = \sigma(y_1, \dots, y_t), \quad (2)$$

which represents all target information observed up to time t . A valid one-step-ahead predictor must satisfy

$$\hat{y}_{t|t-1} \in \mathcal{F}_{t-1}, \quad (3)$$

such that y_t cannot influence its own forecast. The proposed family follows a residual-adaptation decomposition:

$$\hat{y}_{t|t-1} = \hat{y}_{t|t-1}^{\text{base}} + \hat{\Delta}_{t|t-1}, \quad (4)$$

where $\hat{y}_{t|t-1}^{\text{base}}$ is generated by a frozen causal prior and $\hat{\Delta}_{t|t-1}$ is a data-efficient residual correction. The causal base residual is

$$e_t = y_t - \hat{y}_{t|t-1}^{\text{base}}. \quad (5)$$

The proposed models differ only in the representation used to estimate the predictable component of e_t :

$$\text{S3-Forecaster: } e_{1:t-1} \mapsto \mathbf{h}_{t-1} \mapsto \hat{r}_{t|t-1}, \quad (6)$$

$$\text{S3-FastSketch: } e_{1:t-1} \mapsto \phi_{t-1} \mapsto \hat{r}_{t|t-1}. \quad (7)$$

Neither architecture backpropagates through the causal prior or through its nonlinear feature extractor. The learned component is restricted to a regularized linear readout and a low-dimensional calibration mechanism.

2.2. Frozen Causal Forecast Prior

Let

$$f_{\text{base}} : \bigcup_{t \geq 1} \mathbb{R}^t \longrightarrow \mathbb{R} \quad (8)$$

denote a statistical, foundation-model, or heuristic forecasting operator whose parameters are frozen during residual adaptation. Its strictly causal output is

$$\hat{y}_{t|t-1}^{\text{base}} = f_{\text{base}}(y_1, \dots, y_{t-1}; \boldsymbol{\theta}_{\text{base}}). \quad (9)$$

The generic formulation permits the use of rolling estimators, ETS, Prophet, ARIMA, or a zero-shot time-series foundation model. The residual adapter does not alter $\boldsymbol{\theta}_{\text{base}}$. Consequently, adaptation cost is

independent of the number of parameters in the frozen prior. For the lightweight default prior, a causal rolling level estimate is defined as

$$\hat{\mu}_{t-1}^{(w)} = \frac{1}{|\mathcal{W}_{t-1}|} \sum_{j \in \mathcal{W}_{t-1}} y_j, \quad \mathcal{W}_{t-1} = \{\max(1, t-w), \dots, t-1\}. \quad (10)$$

When sufficient monthly history is available, a causal seasonal correction may be added:

$$\hat{s}_{m(t)} = \frac{1}{|\mathcal{I}_{m(t)}|} \sum_{j \in \mathcal{I}_{m(t)}} (y_j - \hat{\mu}_{j-1}^{(w)}), \quad (11)$$

where $\mathcal{I}_{m(t)}$ contains only pre- t observations belonging to the same calendar month as t . The resulting prior is

$$\hat{y}_{t|t-1}^{\text{base}} = \hat{\mu}_{t-1}^{(w)} + \hat{s}_{m(t)}. \quad (12)$$

2.3. S3-Forecaster: Fixed Reservoir Residual Adaptation

2.3.1. Residual Normalization

The residual stream is standardized using statistics estimated exclusively from the training block:

$$\tilde{e}_t = \frac{e_t - \hat{m}_e}{\hat{q}_{0.75}(e) - \hat{q}_{0.25}(e) + \varepsilon_s}, \quad t \in \mathcal{D}_{\text{tr}}, \quad (13)$$

where $\hat{m}_e$ is the training median and $\varepsilon_s > 0$ prevents division by zero. The same frozen transformation is used for calibration and testing.

2.3.2. Echo State Representation

Let R denote the reservoir dimension. The recurrent state is

$$\mathbf{h}_t \in [-1, 1]^R. \quad (14)$$

A leaky ESN update is defined by

$$\tilde{\mathbf{h}}_t = \tanh \left(\mathbf{W}_{\text{in}} \begin{bmatrix} 1 \\ \tilde{e}_t \end{bmatrix} + \mathbf{W}_{\text{res}} \mathbf{h}_{t-1} \right), \quad (15)$$

$$\mathbf{h}_t = (1 - \lambda_{\text{leak}}) \mathbf{h}_{t-1} + \lambda_{\text{leak}} \tilde{\mathbf{h}}_t, \quad \lambda_{\text{leak}} \in (0, 1]. \quad (16)$$

The fixed matrices satisfy

$$\mathbf{W}_{\text{in}} \in \mathbb{R}^{R \times 2}, \quad (17)$$

$$\mathbf{W}_{\text{res}} \in \mathbb{R}^{R \times R}. \quad (18)$$

They are sampled once from seed-controlled random distributions and are not updated during training. Let $\widetilde{\mathbf{W}}_{\text{res}}$ be an initially sampled recurrent matrix. Spectral normalization is performed as

$$\mathbf{W}_{\text{res}} = \rho_{\star} \frac{\widetilde{\mathbf{W}}_{\text{res}}}{\rho(\widetilde{\mathbf{W}}_{\text{res}}) + \varepsilon_{\rho}}, \quad 0 < \rho_{\star} < 1.5, \quad (19)$$

where $\rho(\cdot)$ denotes spectral radius. The stronger contraction condition

$$\|\mathbf{W}_{\text{res}}\|_2 < 1.5 \quad (20)$$

is sufficient for asymptotic forgetting of initial states under a 1-Lipschitz activation. In practice, $\rho(\mathbf{W}_{\text{res}}) < 1.5$ is used as the standard reservoir-design heuristic. Its purpose is to prevent uncontrolled recurrent amplification while retaining a fading memory of the residual history. To supplement the nonlinear reservoir with explicit short-term information, an autoregressive residual vector is constructed:

$$\boldsymbol{\ell}_t^{(p)} = [e_t, e_{t-1}, \dots, e_{t-p+1}]^\top \in \mathbb{R}^p. \quad (21)$$

The complete S3 representation is

$$\mathbf{z}_t = \left[1, \mathbf{h}_t^\top, \left(\boldsymbol{\ell}_t^{(p)} \right)^\top \right]^\top \in \mathbb{R}^P, \quad P = 1 + R + p. \quad (22)$$

2.3.3. Bayesian Ridge Residual Readout

For one-step-ahead forecasting, the readout assumes

$$e_{t+1} = \mathbf{w}^\top \mathbf{z}_t + \xi_{t+1}, \quad \xi_{t+1} \sim \mathcal{N}(0, \beta^{-1}). \quad (23)$$

A zero-mean isotropic Gaussian prior is imposed:

$$p(\mathbf{w} \mid \alpha) = \mathcal{N}(\mathbf{0}, \alpha^{-1} \mathbf{I}_P). \quad (24)$$

Let

$$\mathbf{Z} = [\mathbf{z}_{t_1}, \dots, \mathbf{z}_{t_n}]^\top \in \mathbb{R}^{n \times P}, \quad (25)$$

$$\mathbf{r} = [e_{t_1+1}, \dots, e_{t_n+1}]^\top \in \mathbb{R}^n. \quad (26)$$

The negative log-posterior, excluding constants independent of $\mathbf{w}$, is

$$\mathcal{J}(\mathbf{w}) = \frac{\beta}{2} \|\mathbf{r} - \mathbf{Z}\mathbf{w}\|_2^2 + \frac{\alpha}{2} \|\mathbf{w}\|_2^2. \quad (27)$$

The Hessian is

$$\nabla_{\mathbf{w}}^2 \mathcal{J}(\mathbf{w}) = \beta \mathbf{Z}^\top \mathbf{Z} + \alpha \mathbf{I}_P \succ 0 \quad (28)$$

for $\alpha > 0$. Therefore, the objective is strictly convex and has a unique global minimizer. Setting the gradient to zero gives

$$\nabla_{\mathbf{w}} \mathcal{J}(\mathbf{w}) = -\beta \mathbf{Z}^\top (\mathbf{r} - \mathbf{Z}\mathbf{w}) + \alpha \mathbf{w} = \mathbf{0}, \quad (29)$$

$$(\beta \mathbf{Z}^\top \mathbf{Z} + \alpha \mathbf{I}_P) \mathbf{w} = \beta \mathbf{Z}^\top \mathbf{r}. \quad (30)$$

Hence,

$$\hat{\mathbf{w}}_{\text{MAP}} = (\beta \mathbf{Z}^\top \mathbf{Z} + \alpha \mathbf{I}_P)^{-1} \beta \mathbf{Z}^\top \mathbf{r}. \quad (31)$$

Equivalently, defining $\lambda_R = \alpha/\beta$,

$$\hat{\mathbf{w}}_{\text{MAP}} = (\mathbf{Z}^\top \mathbf{Z} + \lambda_R \mathbf{I}_P)^{-1} \mathbf{Z}^\top \mathbf{r}. \quad (32)$$

The complete posterior is

$$p(\mathbf{w} \mid \mathbf{r}, \mathbf{Z}, \alpha, \beta) = \mathcal{N}(\boldsymbol{\mu}_w, \boldsymbol{\Sigma}_w), \quad (33)$$

$$\boldsymbol{\Sigma}_w = (\alpha \mathbf{I}_P + \beta \mathbf{Z}^\top \mathbf{Z})^{-1}, \quad (34)$$

$$\boldsymbol{\mu}_w = \beta \boldsymbol{\Sigma}_w \mathbf{Z}^\top \mathbf{r}. \quad (35)$$

The one-step residual correction is

$$\hat{r}_{t+1|t} = \boldsymbol{\mu}_w^\top \mathbf{z}_t. \quad (36)$$

The Bayesian derivation provides a probabilistic interpretation of the regularized readout. When fixed precision parameters are used, its posterior mode is numerically equivalent to conventional Ridge regression. Lasso and Elastic Net may be considered as convex sensitivity variants, but they must not be described as Bayesian Ridge unless the corresponding prior interpretation is explicitly introduced.

2.3.4. Volatility-Sensitive OOB Gate

The residual correction is not assumed to be uniformly reliable. A causal volatility process is therefore computed from the prior residuals. Let

$$\mu_t^{(e)} = (1 - \lambda_v) \mu_{t-1}^{(e)} + \lambda_v e_t, \quad (37)$$

$$q_t^{(e)} = (1 - \lambda_v) q_{t-1}^{(e)} + \lambda_v \left(e_t - \mu_{t-1}^{(e)} \right)^2, \quad (38)$$

$$v_t = \sqrt{q_t^{(e)} + \varepsilon_v}. \quad (39)$$

The volatility is robustly standardized using training-only statistics:

$$\tilde{v}_t = \frac{v_t - \text{median}_{j \in \mathcal{D}_{\text{tr}}}(v_j)}{\text{IQR}_{j \in \mathcal{D}_{\text{tr}}}(v_j)/1.349 + \varepsilon_v}. \quad (40)$$

The gate is

$$g_t(a, b) = \sigma(a(\tilde{v}_t - b)) = \frac{1}{1 + \exp[-a(\tilde{v}_t - b)]}, \quad g_t \in (0, 1). \quad (41)$$

The ESN and its readout are fitted on $\mathcal{D}_{\text{tr}}$. Keeping these components fixed, the two gate parameters are estimated exclusively on the chronological calibration block:

$$(\hat{a}, \hat{b}) = \arg \min_{a \in [a_{\min}, a_{\max}], b \in [b_{\min}, b_{\max}]} \frac{1}{|\mathcal{D}_{\text{cal}}|} \sum_{t \in \mathcal{D}_{\text{cal}}} |e_t - g_{t-1}(a, b) \hat{r}_{t|t-1}|. \quad (42)$$

This is an out-of-bag optimization with respect to the readout because no calibration observation is used to estimate $\mathbf{w}$. The gate is a two-parameter calibration mechanism rather than a second high-capacity predictor. The final S3 point forecast is

$$\hat{y}_{t|t-1}^{\text{S3}} = \hat{y}_{t|t-1}^{\text{base}} + g_{t-1}(\hat{a}, \hat{b}) \hat{r}_{t|t-1}. \quad (43)$$

The term *gradient-free* refers to the absence of end-to-end backpropagation through the causal prior and reservoir. The bounded low-dimensional calibration of (a, b) is not conflated with neural gradient-based fine-tuning.

2.4. S3-FastSketch: Lightweight Multiscale Residual Adaptation

2.4.1. Causal Multiscale Feature Map

S3-FastSketch replaces the recurrent reservoir by a deterministic causal map

$$\Phi : e_{1:t} \mapsto \phi_t \in \mathbb{R}^{D_\phi}. \quad (44)$$

The representation is

$$\phi_t = [\ell_t^\top, \mathbf{a}_t^\top, \mathbf{v}_t^\top, \mathbf{d}_t^\top, \mathbf{c}_t^\top, \mathbf{s}_t^\top]^\top. \quad (45)$$

Let $\mathcal{Q} = \{q_1, \dots, q_Q\}$ be a set of causal smoothing spans and $\mathcal{K} = \{k_1, \dots, k_K\}$ a set of local filter scales.

Residual lag block.. For p autoregressive lags,

$$\ell_t = [e_{t-1}, e_{t-2}, \dots, e_{t-p}]^\top \in \mathbb{R}^p. \quad (46)$$

Exponentially weighted residual block.. For each $q \in \mathcal{Q}$,

$$a_t^{(q)} = (1 - \eta_q) a_{t-1}^{(q)} + \eta_q e_t, \quad (47)$$

$$\eta_q = \frac{2}{q+1}. \quad (48)$$

The block is

$$\mathbf{a}_t = [a_t^{(q_1)}, \dots, a_t^{(q_Q)}]^\top. \quad (49)$$

Residual-magnitude block.. The multiscale magnitude summaries are

$$v_t^{(q)} = (1 - \eta_q) v_{t-1}^{(q)} + \eta_q |e_t|, \quad (50)$$

and

$$\mathbf{v}_t = [v_t^{(q_1)}, \dots, v_t^{(q_Q)}]^\top. \quad (51)$$

Local-difference block.. For $q \in \mathcal{Q}$,

$$d_t^{(q)} = \begin{cases} e_t - e_{t-q}, & t > q, \\ 0, & t \leq q. \end{cases} \quad (52)$$

Thus,

$$\mathbf{d}_t = [d_t^{(q_1)}, \dots, d_t^{(q_Q)}]^\top. \quad (53)$$

Fixed causal convolutional block.. For each $k \in \mathcal{K}$, define the causal window

$$\mathcal{W}_{t,k} = \{\max(1, t - k + 1), \dots, t\}. \quad (54)$$

Its local mean is

$$c_{t,k}^{(\text{mean})} = \frac{1}{|\mathcal{W}_{t,k}|} \sum_{j \in \mathcal{W}_{t,k}} e_j. \quad (55)$$

Its endpoint slope is

$$c_{t,k}^{(\text{slope})} = e_t - e_{\max(1, t-k+1)}. \quad (56)$$

Let $\mathcal{W}_{t,k}^L$ and $\mathcal{W}_{t,k}^R$ denote the first and second halves of $\mathcal{W}_{t,k}$. The local contrast is

$$c_{t,k}^{(\text{contrast})} = \frac{1}{|\mathcal{W}_{t,k}^R|} \sum_{j \in \mathcal{W}_{t,k}^R} e_j - \frac{1}{|\mathcal{W}_{t,k}^L|} \sum_{j \in \mathcal{W}_{t,k}^L} e_j. \quad (57)$$

The local absolute energy is

$$c_{t,k}^{(\text{energy})} = \frac{1}{|\mathcal{W}_{t,k}|} \sum_{j \in \mathcal{W}_{t,k}} |e_j|. \quad (58)$$

The convolutional block is the concatenation

$$\mathbf{c}_t = \text{concat}_{k \in \mathcal{K}} \left[c_{t,k}^{(\text{mean})}, c_{t,k}^{(\text{slope})}, c_{t,k}^{(\text{contrast})}, c_{t,k}^{(\text{energy})} \right]. \quad (59)$$

Seasonal and calendar block.. For monthly seasonality $m = 12$,

$$\mathbf{s}_t = \left[e_{t-m}, e_t - e_{t-m}, \sin\left(\frac{2\pi t}{m}\right), \cos\left(\frac{2\pi t}{m}\right) \right]^\top, \quad (60)$$

where unavailable seasonal lags are initialized to zero. Each feature dimension is standardized using training medians and interquartile ranges:

$$\tilde{\phi}_{t,j} = \frac{\phi_{t,j} - \text{median}_{u \in \mathcal{D}_{\text{tr}}}(\phi_{u,j})}{\text{IQR}_{u \in \mathcal{D}_{\text{tr}}}(\phi_{u,j})/1.349 + \varepsilon_\phi}. \quad (61)$$

2.4.2. Regularized Residual Readout

For a forecast horizon H , define

$$\mathbf{r}_{t+1:t+H} = [e_{t+1}, \dots, e_{t+H}]^\top \in \mathbb{R}^H. \quad (62)$$

The multivariate residual readout is

$$\hat{\mathbf{r}}_{t+1:t+H|t} = \mathbf{W}^\top \tilde{\boldsymbol{\phi}}_t, \quad \mathbf{W} \in \mathbb{R}^{D_\phi \times H}. \quad (63)$$

Let

$$\boldsymbol{\Phi} \in \mathbb{R}^{n_{\text{tr}} \times D_\phi}, \quad (64)$$

$$\mathbf{R} \in \mathbb{R}^{n_{\text{tr}} \times H} \quad (65)$$

contain the training sketches and future residual targets. The readout solves

$$\widehat{\mathbf{W}} = \arg \min_{\mathbf{W}} \|\mathbf{R} - \boldsymbol{\Phi} \mathbf{W}\|_F^2 + \lambda_F \|\mathbf{W}\|_F^2. \quad (66)$$

Its closed-form solution is

$$\widehat{\mathbf{W}} = (\boldsymbol{\Phi}^\top \boldsymbol{\Phi} + \lambda_F \mathbf{I}_{D_\phi})^{-1} \boldsymbol{\Phi}^\top \mathbf{R}. \quad (67)$$

2.4.3. Projected Calibration Shrinkage

The raw residual forecast may be unstable when the calibration block is small. For each horizon h , a single bounded coefficient is therefore estimated from calibration pairs

$$\{(r_{i,h}, \hat{r}_{i,h})\}_{i \in \mathcal{D}_{\text{cal}}} . \quad (68)$$

The calibration objective is

$$\mathcal{L}_h(\gamma) = \sum_{i \in \mathcal{D}_{\text{cal}}} (r_{i,h} - \gamma \hat{r}_{i,h})^2 . \quad (69)$$

Differentiating gives

$$\frac{\partial \mathcal{L}_h}{\partial \gamma} = -2 \sum_{i \in \mathcal{D}_{\text{cal}}} \hat{r}_{i,h} (r_{i,h} - \gamma \hat{r}_{i,h}) , \quad (70)$$

$$0 = \sum_{i \in \mathcal{D}_{\text{cal}}} \hat{r}_{i,h} r_{i,h} - \gamma \sum_{i \in \mathcal{D}_{\text{cal}}} \hat{r}_{i,h}^2 . \quad (71)$$

The unconstrained solution is

$$\tilde{\gamma}_h = \frac{\sum_{i \in \mathcal{D}_{\text{cal}}} \hat{r}_{i,h} r_{i,h}}{\sum_{i \in \mathcal{D}_{\text{cal}}} \hat{r}_{i,h}^2 + \varepsilon_\gamma} . \quad (72)$$

The final coefficient is

$$\hat{\gamma}_h = \Pi_{[0, \gamma_{\text{max}}]}(\tilde{\gamma}_h) , \quad (73)$$

where the Euclidean projection is

$$\Pi_{[0, \gamma_{\text{max}}]}(x) = \min \{ \gamma_{\text{max}}, \max \{ 0, x \} \} . \quad (74)$$

The lower boundary suppresses a residual correction whose calibration association is non-positive, whereas the upper boundary prevents excessive amplification. Only one scalar is estimated per horizon, thereby limiting calibration variance without requiring nonlinear gate optimization. The FastSketch forecast is

$$\hat{y}_{t+h|t}^{\text{FS}} = \hat{y}_{t+h|t}^{\text{base}} + \hat{\gamma}_h \hat{r}_{t+h|t} , \quad h = 1, \dots, H . \quad (75)$$

The reported experiments use $H = 1$. The multi-horizon notation is retained to show that the architecture extends directly by assigning an independent readout column and shrinkage coefficient to each horizon.

2.5. Adaptive Conformal Uncertainty Estimation

The same uncertainty-estimation layer is applied after either S3-Forecaster or S3-FastSketch. Let $\hat{y}_{t|t-1}$ denote the final point forecast. The nonconformity score is

$$s_t = |y_t - \hat{y}_{t|t-1}| . \quad (76)$$

For $H > 1$, a simultaneous score can be defined as

$$s_t^{(H)} = \max_{h=1, \dots, H} |y_{t+h} - \hat{y}_{t+h|t}| . \quad (77)$$

2.5.1. Dynamic Score Buffer

Let W denote the maximum calibration-buffer length. Immediately before forecasting y_t , the score buffer is

$$\mathcal{S}_t = \{s_j : \max(t - W, t_{\text{cal},1}) \leq j \leq t - 1\}, \quad (78)$$

where $t_{\text{cal},1}$ is the first calibration index. Thus,

$$|\mathcal{S}_t| \leq W. \quad (79)$$

Given an effective miscoverage level δ_t , define

$$k_t = \min \{|\mathcal{S}_t|, \lceil (|\mathcal{S}_t| + 1)(1 - \delta_t) \rceil\}. \quad (80)$$

The finite-buffer empirical quantile is

$$q_t(\delta_t) = s_{(k_t)}(\mathcal{S}_t), \quad (81)$$

where $s_{(k_t)}(\mathcal{S}_t)$ is the k_t -th order statistic. The adaptive conformal interval is

$$\widehat{C}_t = [\widehat{y}_{t|t-1} - q_t(\delta_t), \widehat{y}_{t|t-1} + q_t(\delta_t)]. \quad (82)$$

2.5.2. Sequential Miscoverage Update

After observing y_t , define

$$M_t = \mathbb{I}\{y_t \notin \widehat{C}_t\}. \quad (83)$$

Let δ^* be the target miscoverage, with $\delta^* = 0.10$ for nominal 90% coverage. The effective miscoverage is updated as

$$\delta_{t+1} = \Pi_{[\delta_{\min}, \delta_{\max}]} [\delta_t + \eta_{\text{ACI}} (\delta^* - M_t)], \quad (84)$$

where

$$\Pi_{[\delta_{\min}, \delta_{\max}]}(x) = \min\{\delta_{\max}, \max\{\delta_{\min}, x\}\}. \quad (85)$$

When a miss occurs, $M_t = 1$, the update decreases δ_{t+1} and therefore raises the next empirical quantile, producing a wider interval. When coverage is repeatedly achieved, δ_t can increase gradually, reducing excessive width. Finally,

$$\mathcal{S}_{t+1} = \text{Tail}_W(\mathcal{S}_t \cup \{s_t\}). \quad (86)$$

This construction does not assume a Gaussian residual distribution. Its distribution-free interpretation concerns the absence of a parametric error law under conformal validity conditions. Under serial dependence and temporal drift, ACI targets adaptive long-run miscoverage control; it does not imply exact conditional coverage for every individual forecast or every short test block.

2.6. Model Complexity

For S3-Forecaster, the number of learned readout coefficients is approximately

$$P_{S3} = H(1 + R + p) + 2, \quad (87)$$

where the final two parameters correspond to the OOB gate. The fixed matrices $\mathbf{W}_{\text{in}}$ and $\mathbf{W}_{\text{res}}$ are excluded because they are not trained.

For S3-FastSketch,

$$P_{\text{FS}} = HD_{\phi} + H, \quad (88)$$

where the final H coefficients are the calibration shrinkage factors. For a dense reservoir, state propagation requires

$$\mathcal{O}(NR^2) \quad (89)$$

operations, which can be reduced using a sparse recurrent matrix. FastSketch feature construction requires approximately

$$\mathcal{O}(ND_{\phi}). \quad (90)$$

Both methods avoid iterative backpropagation through a nonlinear sequence model. Their learned readouts are obtained through convex regularized objectives, while FastSketch additionally avoids nonlinear gate calibration.

3. Experimental Setup

3.1. Datasets

The evaluation comprises one proprietary engineering application and four public monthly forecasting collections. All evaluated series satisfy $N < 200$ and preserve their original chronological order.

3.1.1. Industrial Construction-Materials Demand

The primary engineering application is an anonymized *Industrial Construction-Materials Demand* dataset, denoted ICMD. The source database contains 61,395 transactional records, 47 original variables, 555 product references, and 62 monthly periods between January 2020 and February 2025.

The database is sparse at product level: many references are active during only a limited subset of months. This characteristic makes global high-capacity models difficult to estimate for individual products and motivates an architecture with controlled trainable complexity. The target series was constructed by aggregating positive sales value for a selected construction-material reference:

$$y_t = \sum_{j \in \mathcal{T}_t} \text{SalesValue}_j, \quad (91)$$

where $\mathcal{T}_t$ is the set of valid transactions assigned to month t . The evaluated contiguous series contains 48 monthly observations from January 2020 to December 2023. The company identity and product identifier are withheld for confidentiality, whereas the target definition, aggregation procedure, temporal split, and model implementation remain fully specified.

Table 1: Evaluated time series and chronological partitions.

Dataset	Domain	N	Train	Calibration	Test
ICMD	Industrial sales	48	30	8	10
CIF 2016	Banking	60	38	10	12
M3 Monthly	Competition series	90	57	15	18
Tourism Monthly	Tourism demand	120	76	20	24
M4 Monthly	Competition series	90	57	15	18

3.1.2. Public Benchmark Series

The public evaluation includes:

- **M4 Monthly**: a monthly series drawn from the M4 forecasting competition, used as the principal reproducible methodological validation [45];
- **M3 Monthly**: a monthly series from the M3 forecasting competition [46];
- **Tourism Monthly**: a monthly tourism-demand series from the Tourism Forecasting Competition [47];
- **CIF 2016**: a monthly banking-domain series from the Computational Intelligence in Forecasting competition [48].

A fixed random seed and predefined eligibility rules were used for public-series selection. Eligibility required monthly frequency, finite observations, a minimum length sufficient for the chronological split, and a final history length below 200. Series selection was completed before model evaluation and was not conditioned on forecasting performance.

3.1.3. Preprocessing and Data Integrity

All series were converted to a regular monthly index and sorted chronologically. Duplicate timestamps were aggregated before splitting. Transformations requiring estimated statistics were fitted exclusively on the training block and then reused without re-estimation on calibration and test observations. Let $\mathcal{T}_{\text{prep}}$ denote any scaling or variance-stabilizing transformation. The causal preprocessing requirement is

$$\mathcal{T}_{\text{prep}} = \mathcal{T}_{\text{prep}}(\{y_t : t \in \mathcal{D}_{\text{tr}}\}), \quad (92)$$

and not

$$\mathcal{T}_{\text{prep}} = \mathcal{T}_{\text{prep}}(\{y_t : t \in \mathcal{D}_{\text{tr}} \cup \mathcal{D}_{\text{cal}} \cup \mathcal{D}_{\text{te}}\}). \quad (93)$$

Forecasts were inverse-transformed before computing the primary test metrics, so all benchmark values are reported on the original application scale.

3.2. Baseline Models

The proposed methods are compared with ten reference models spanning five forecasting paradigms.

3.2.1. Statistical Models

1. **AutoRegressive model (AR)**. A finite lag model

$$y_t = c + \sum_{j=1}^p \varphi_j y_{t-j} + \epsilon_t \quad (94)$$

provides a low-complexity linear benchmark.

2. **ARIMA**. The model satisfies

$$\Phi(B)(1 - B)^d y_t = c + \Theta(B)\epsilon_t, \quad (95)$$

where B is the lag operator and (p, d, q) is selected from a bounded dataset-level candidate set.

3. **ETS**. Exponential smoothing is evaluated using combinations of additive error, trend, damping, and monthly seasonal components. Seasonal terms are disabled when the available history is insufficient for reliable estimation.

3.2.2. Classical Machine-Learning Models

4. **Kernel Ridge Regression (KRR)**. Lag windows are mapped through a kernel $k(\mathbf{x}_i, \mathbf{x}_j)$ and fitted using dual Ridge regularization.
5. **Gaussian Process Regression (GP)**. Direct lag-to-horizon mappings are estimated with RBF or Matérn covariance functions and an additive white-noise component.

3.2.3. Deep-Learning Models

6. **Long Short-Term Memory network (LSTM)**. One- or two-layer recurrent architectures are trained on chronological lag windows with early stopping and no shuffling.
7. **Causal one-dimensional CNN**. Causal convolutional layers extract local temporal patterns, followed by a dense direct-forecast head.

3.2.4. Modern Lightweight and KAN Architectures

8. **AR-KAN**. An autoregressive lag representation is processed by a Kolmogorov–Arnold Network with learnable spline-based transformations.
9. **EasyTSF/NLinear**. A lightweight linear forecasting model predicts deviations from the last observed value:

$$\hat{y}_{t+1} = y_t + \mathbf{w}^\top (\mathbf{x}_t - y_t \mathbf{1}). \quad (96)$$

3.2.5. Time-Series Foundation Model

10. **Chronos**. Chronos is evaluated in zero-shot mode with a frozen pretrained checkpoint and without gradient-based adaptation to the target series [11].

Table 2: Dataset-level hyperparameter families.

Model	Dataset-level configuration dimensions
S3-Forecaster	Reservoir size, target spectral radius, residual lags, prior window, Bayesian/Ridge regularization, gate bounds, ACI step size, and calibration buffer length.
S3-FastSketch	Residual lags, EMA spans, convolutional scales, seasonal/calendar features, Ridge regularization, maximum shrinkage, and ACI step size.
ARIMA	Orders (p, d, q) and deterministic trend specification.
AR	Lag order, trend terms, and optional seasonal indicators.
ETS	Error, trend, seasonal, and damped-trend components.
KRR	Input window, kernel, regularization, kernel scale, and polynomial parameters.
GP	Input window, kernel family, length scale, signal magnitude, and noise level.
LSTM	Input window, number of layers, hidden units, dropout, learning rate, batch size, and training budget.
CNN	Input window, filters, kernel width, dense width, dropout, learning rate, and training budget.
AR-KAN	Autoregressive window, hidden width, spline grid, spline degree, regularization, and training budget.
EasyTSF	NLinear/DLinear architecture, window length, decomposition kernel, learning rate, weight decay, and training budget.
Chronos	Frozen checkpoint, context length, sampling configuration, and output quantiles; no target-domain gradient optimization.

3.3. Offline Hyperparameter Selection and Frozen Operational Reuse

Hyperparameter selection and operational evaluation are treated as two separate computational phases. For model m and dataset d , let

$$\mathcal{G}_{m,d} = \left\{ \boldsymbol{\theta}_{m,d}^{(1)}, \dots, \boldsymbol{\theta}_{m,d}^{(K)} \right\} \quad (97)$$

denote a finite dataset-level micro-grid. The selected configuration is

$$\boldsymbol{\theta}_{m,d}^* = \arg \min_{\boldsymbol{\theta} \in \mathcal{G}_{m,d}} \frac{1}{|\mathcal{I}_d^{\text{dev}}|} \sum_{i \in \mathcal{I}_d^{\text{dev}}} \mathcal{L}_{\text{val}} \left(m, \boldsymbol{\theta}; \mathbf{y}^{(i)} \right), \quad (98)$$

where $\mathcal{I}_d^{\text{dev}}$ contains only development series or pre-test development partitions from dataset d . Point-forecast configurations are selected using validation MAPE_e . Uncertainty parameters are selected using

$$\mathcal{J}_{\text{UQ}} = \text{MSIS} + \lambda_{\text{cov}} [\max(0, (1 - \delta^*) - \text{ECP})]^2. \quad (99)$$

After selection, $\boldsymbol{\theta}_{m,d}^*$ is serialized and reused without modification in the final benchmark, ablation, data-efficiency, and multi-prior experiments. No test observation contributes to Eq. (98). The principal configuration dimensions are summarized in Table 2.

The execution times reported in the principal benchmark correspond to

$$t_{\text{operational}} = t_{\text{fit}} + t_{\text{forecast}}, \quad (100)$$

after loading the frozen dataset-level configuration. The one-time HPO cost is reported separately and is not conflated with recurring deployment latency.

3.4. Chronological Validation and Rolling Evaluation

For a series of length N , define

$$n_{\text{tr}} = \lfloor 0.64N \rfloor, \quad (101)$$

$$n_{\text{dev}} = \lfloor 0.80N \rfloor, \quad (102)$$

$$n_{\text{cal}} = n_{\text{dev}} - n_{\text{tr}}, \quad (103)$$

$$n_{\text{te}} = N - n_{\text{dev}}. \quad (104)$$

The chronological blocks are

$$\mathcal{D}_{\text{tr}} = \{1, \dots, n_{\text{tr}}\}, \quad (105)$$

$$\mathcal{D}_{\text{cal}} = \{n_{\text{tr}} + 1, \dots, n_{\text{dev}}\}, \quad (106)$$

$$\mathcal{D}_{\text{te}} = \{n_{\text{dev}} + 1, \dots, N\}. \quad (107)$$

The readout parameters are estimated only from $\mathcal{D}_{\text{tr}}$. The OOB gate, FastSketch shrinkage, and initial conformity-score buffer are estimated only from $\mathcal{D}_{\text{cal}}$. Test observations are reserved for final rolling evaluation. For every $t \in \mathcal{D}_{\text{te}}$, the model produces

$$\left(\hat{y}_{t|t-1}, \hat{L}_t, \hat{U}_t \right) = \mathcal{A} \left(y_{1:t-1}; \boldsymbol{\theta}_{m,d}^* \right), \quad (108)$$

with horizon $H = 1$ and rolling step equal to one. The causality requirement is

$$\hat{y}_{t|t-1}, \hat{L}_t, \hat{U}_t \in \mathcal{F}_{t-1}. \quad (109)$$

After the prediction is recorded, y_t is revealed and used only to compute the test error, to compute the nonconformity score s_t , and to update the ACI state for time $t + 1$. Under the additional requirement that all preprocessing parameters are fitted on $\mathcal{D}_{\text{tr}}$ only, Eqs. (108)–(109) exclude look-ahead bias by construction. The readout and gate remain frozen throughout $\mathcal{D}_{\text{te}}$. Only the causal state, prior history, and ACI buffer are updated sequentially.

3.5. Prediction Intervals for Reference Models

For models that provide native predictive quantiles, including probabilistic ARIMA, ETS, GP, and Chronos configurations, the corresponding nominal 90% intervals are used. For deterministic baselines, a common post-hoc interval is constructed from calibration residuals:

$$u_t = y_t - \hat{y}_{t|t-1}, \quad t \in \mathcal{D}_{\text{cal}}. \quad (110)$$

The calibration standard deviation is

$$\hat{\sigma}_{\text{cal}} = \sqrt{\frac{1}{|\mathcal{D}_{\text{cal}}| - 1} \sum_{t \in \mathcal{D}_{\text{cal}}} (u_t - \bar{u}_{\text{cal}})^2}. \quad (111)$$

The resulting reference interval is

$$\hat{C}_t^{\text{Gauss}} = [\hat{y}_{t|t-1} - z_{1-\delta/2} \hat{\sigma}_{\text{cal}}, \hat{y}_{t|t-1} + z_{1-\delta/2} \hat{\sigma}_{\text{cal}}], \quad (112)$$

where $z_{0.95} \approx 1.645$ for $\delta = 0.10$. This construction enables a common interval-score comparison, but it is not presented as theoretically equivalent to adaptive conformal inference.

3.6. Evaluation Metrics and Protocol

Let

$$\left\{ (y_t, \hat{y}_t, \hat{L}_t, \hat{U}_t) \right\}_{t=1}^T \quad (113)$$

denote the forecasts and intervals over the final test block.

3.6.1. Mean Squared Error

$$\text{MSE} = \frac{1}{T} \sum_{t=1}^T (y_t - \hat{y}_t)^2. \quad (114)$$

3.6.2. Root Mean Squared Error

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{t=1}^T (y_t - \hat{y}_t)^2}. \quad (115)$$

3.6.3. Coefficient of Determination

Let

$$\bar{y} = \frac{1}{T} \sum_{t=1}^T y_t. \quad (116)$$

Then

$$R^2 = 1 - \frac{\sum_{t=1}^T (y_t - \hat{y}_t)^2}{\sum_{t=1}^T (y_t - \bar{y})^2}. \quad (117)$$

When the test variance is zero, R^2 is undefined and is not used for ranking.

3.6.4. Epsilon-Stabilized MAPE

To prevent division by zero or numerical instability near zero-valued targets, we define

$$\text{MAPE}_\varepsilon = \frac{100}{T} \sum_{t=1}^T \frac{|y_t - \hat{y}_t|}{\max(|y_t|, \varepsilon)}, \quad \varepsilon = 1 \times 10^{-5}. \quad (118)$$

The factor 100 expresses the metric as a percentage.

3.6.5. Empirical Coverage Probability

At nominal coverage $1 - \delta$,

$$\text{ECP}(1 - \delta) = \frac{1}{T} \sum_{t=1}^T \mathbb{I} \left\{ y_t \in [\hat{L}_t, \hat{U}_t] \right\}. \quad (119)$$

Coverage quality is evaluated through deviation from the target:

$$\Delta_{\text{cov}} = |\text{ECP}(1 - \delta) - (1 - \delta)|. \quad (120)$$

Coverage values above the target are not automatically interpreted as superior, because they may result from excessively wide intervals.

3.6.6. Interval Width

$$\text{IW} = \frac{1}{T} \sum_{t=1}^T (\hat{U}_t - \hat{L}_t). \quad (121)$$

3.6.7. Interval Score

For miscoverage δ , the interval score is

$$\begin{aligned} \text{IS}_{\delta,t} = & (\hat{U}_t - \hat{L}_t) \\ & + \frac{2}{\delta} (\hat{L}_t - y_t) \mathbb{I} \left\{ y_t < \hat{L}_t \right\} \\ & + \frac{2}{\delta} (y_t - \hat{U}_t) \mathbb{I} \left\{ y_t > \hat{U}_t \right\}. \end{aligned} \quad (122)$$

The first term penalizes width, whereas the remaining terms penalize misses below and above the interval.

3.6.8. Seasonal Scaling Factor

For monthly seasonality $m = 12$, the in-sample seasonal-naive scale is

$$s_m = \frac{1}{n_{\text{tr}} - m} \sum_{t=m+1}^{n_{\text{tr}}} |y_t - y_{t-m}|. \quad (123)$$

A numerical safeguard is applied as

$$\tilde{s}_m = \max(s_m, \varepsilon_s). \quad (124)$$

3.6.9. Mean Scaled Interval Score

The MSIS is

$$\text{MSIS}_{\delta} = \frac{1}{T} \sum_{t=1}^T \frac{\text{IS}_{\delta,t}}{\tilde{s}_m}. \quad (125)$$

Lower MSIS indicates a more favorable joint balance between interval sharpness and coverage.

3.6.10. Operational Runtime and Trainable Parameters

Operational runtime is

$$\text{Time}_{\text{op}} = t_{\text{end}} - t_{\text{start}}, \quad (126)$$

where timing begins immediately before final fitting with the stored configuration and ends after the test forecasts are produced. Trainable-parameter counts include coefficients estimated from the target series but exclude frozen foundation-model parameters, fixed ESN matrices, and fixed FastSketch filters. Accordingly, parameter count measures target-domain statistical complexity rather than total memory footprint.

3.7. Reproducibility Controls

The following controls are applied throughout the experiments:

1. all dataset and model random seeds are fixed and recorded;
2. public-series identifiers and truncation lengths are stored in configuration files;
3. dataset-level HPO configurations are serialized before final testing;
4. no test observation is used for feature scaling, model selection, gate calibration, or initialization of the conformal buffer;
5. neural models are trained without temporal shuffling and with validation-based early stopping;
6. point forecasts and uncertainty intervals are evaluated on the original data scale;
7. operational runtime and one-time offline HPO cost are reported separately.

This protocol ensures that the empirical comparison evaluates frozen dataset-level configurations under a common strictly causal forecasting workflow, while preserving the distinction between predictive accuracy, adaptive conformal uncertainty estimation, and operational computational cost.

4. Results and Discussion

4.1. Engineering Case Characterization and Non-Stationarity

The primary engineering application is an anonymized *Industrial Construction-Materials Demand* dataset, hereafter ICMD. The source database contains heterogeneous transactional records rather than a pre-assembled forecasting benchmark. It comprises 61,395 records, 47 variables, 555 product references, and 62 monthly periods between January 2020 and February 2025. Product-level histories are markedly sparse: the median product is observed in only approximately six monthly periods. This catalog-level sparsity is operationally relevant because it prevents the routine use of high-capacity models for most references and motivates a forecasting framework whose trainable complexity remains controlled when $N < 200$.

The target was constructed by chronologically aggregating the positive sales value of the selected construction-material reference at monthly frequency. The evaluated contiguous series contains 48 observations from January 2020 to December 2023, of which the last 10 months constitute the final test block.

Table 3: Industrial dataset profile.

Characteristic	Value
Transactional records	61,395
Original variables	47
Product references	555
Database coverage	Jan. 2020–Feb. 2025
Monthly periods in database	62
Median active months per product	≈ 6
Evaluated series coverage	Jan. 2020–Dec. 2023
Evaluated monthly observations	48
Final test observations	10
Forecast target	Monthly aggregated sales value

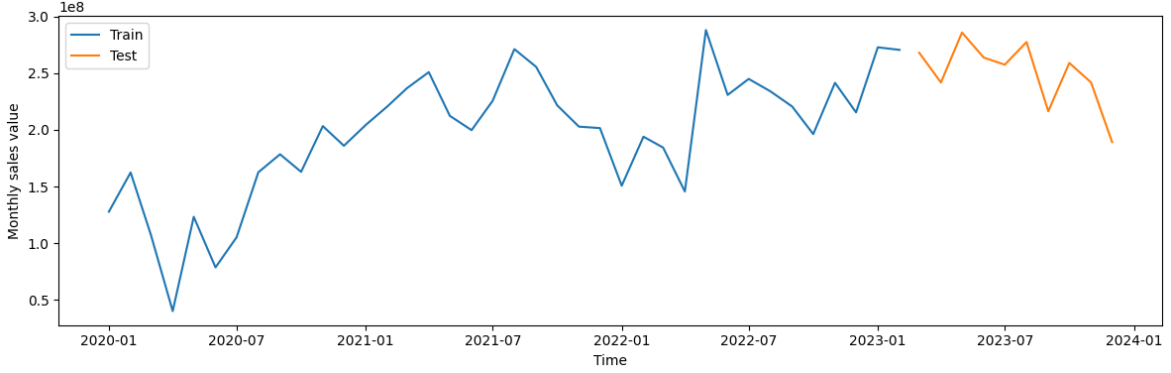


Figure 1: Industrial demand series and temporal split.

The company identity and product identifier are omitted for confidentiality, while the aggregation rule, temporal split, target definition, and model code remain fully specified. Table 3 summarizes the resulting data audit.

Figure 1 shows a changing level and substantial short-range variation throughout the observed window. The dominant empirical property is therefore non-stationarity in location and local scale, rather than the presence of a threshold-defined extreme event in the final test period. This distinction is central to the present interpretation: the experiments evaluate adaptation under short, non-stationary histories, but do not claim test-time evidence for abrupt structural discontinuities.

Because the anonymized release does not expose the raw product-level industrial series, the diagnostic values in Table 4 are reported as audit fields rather than inferred from forecast outputs. They should be computed directly from the confidential monthly target before final submission; until then, the interpretation below is restricted to the visible series profile and benchmark outcomes.

Table 4: Industrial series diagnostics.

Diagnostic	Raw scale	Log scale
ADF p -value	<i>to be audited</i>	<i>to be audited</i>
KPSS p -value	<i>to be audited</i>	<i>to be audited</i>
Coefficient of variation	<i>to be audited</i>	<i>to be audited</i>
Lag-1 autocorrelation	<i>to be audited</i>	<i>to be audited</i>
Lag-12 autocorrelation	<i>to be audited</i>	<i>to be audited</i>
Seasonal strength	<i>to be audited</i>	<i>to be audited</i>
Trend strength	<i>to be audited</i>	<i>to be audited</i>

4.2. Overall Benchmarking and Metric-Specific Behaviour

Table 5 preserves all five datasets and all model evaluations. Each row represents a dataset–model pair. ICMD is treated as the primary engineering validation, whereas M4 provides the principal public and reproducible methodological validation. CIF, M3, and Tourism are retained to expose both transferability and failure modes.

On ICMD, S3-Forecaster obtains the strongest scale-normalized RMSE and R^2 , whereas S3-FastSketch gives the lowest MAPE_ϵ . Chronos is highly competitive—second in RMSE and R^2 and close in MAPE_ϵ —but it does not dominate all point metrics on the model scale. The engineering case therefore supports a more nuanced conclusion: a frozen foundation prior can be strong, but gradient-free residual adaptation remains competitive and can provide the best variance-sensitive accuracy under an extremely short industrial history.

On M4, S3-FastSketch obtains the best point-forecasting results across RMSE, R^2 , and MAPE_ϵ , while S3-Forecaster is consistently second. This pattern indicates that the lower-dimensional sketch can be preferable when the residual structure is sufficiently captured by compact random features. It also shows that the S3 family benefit is not tied to a single residual representation.

CIF and M3 delimit the method’s operating range. On CIF, S3-Forecaster is the best point model across all three metrics, with S3-FastSketch and Gaussian Process remaining competitive. On M3, S3-Forecaster and ETS are tied at the best point values, while Chronos is the strongest non-S3/non-ETS competitor in RMSE and R^2 . Tourism provides a complementary case in which S3-FastSketch is best and EasyTSF is second, showing that compact adaptation may outperform the larger reservoir when the available history supports a simpler correction. Overall, Table 5 supports selective S3-family strength rather than universal dominance by a single model.

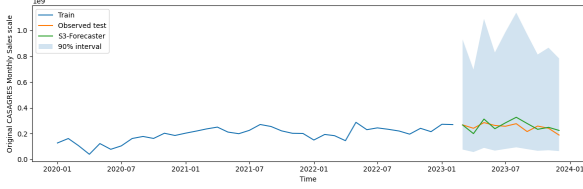
Best values are shown in bold and second-best values are underlined within each dataset and metric.

4.3. Component-Wise Ablation Study

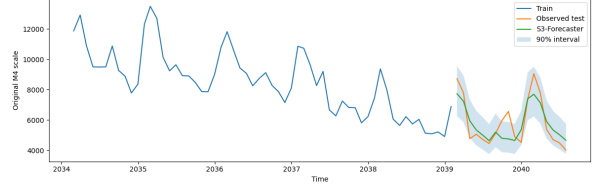
Table 6 separates the frozen causal prior, the ESN-based residual adapter, the OOB volatility gate, and the adaptive conformal uncertainty layer. The residual adapter has conditional rather than uniform utility. Let e_t denote the base residual and $\hat{r}_t$ the adapter correction. For a gate value g_t , the change in expected

Table 5: Point-forecasting results on the model scale.

Dataset	Model	RMSE $\downarrow$	R^2 $\uparrow$	MAPE $_{\epsilon}$ $\downarrow$
CIF	S3-Forecaster	0.0161	0.5014	0.2103
	S3-FastSketch	0.0174	0.4163	0.2312
	ARIMA	0.0251	-0.2110	0.2740
	AR	0.0323	-1.0104	0.3908
	ETS	0.0225	0.0282	0.2446
	Kernel Ridge	0.0217	0.0925	0.2569
	Gaussian Process	0.0186	0.3381	0.2179
	Chronos	0.0205	0.1891	0.2303
	LSTM	0.0654	-7.2221	0.7513
	CNN	0.0319	-0.9553	0.3323
	AR-KAN	0.0605	-6.0433	0.7468
	EasyTSF	0.0232	-0.0337	0.2550
M3	S3-Forecaster	0.3773	0.0176	3.1504
	S3-FastSketch	0.3952	-0.0777	3.2284
	ARIMA	0.6076	-1.5471	5.9327
	AR	0.4512	-0.4051	3.7083
	ETS	0.3773	0.0176	3.1504
	Kernel Ridge	0.7047	-2.4268	7.3793
	Gaussian Process	0.5155	-0.8339	5.3484
	Chronos	0.3925	-0.0631	3.2677
	LSTM	0.4585	-0.4508	4.2237
	CNN	0.4849	-0.6224	4.7572
	AR-KAN	0.4272	-0.2592	4.1712
	EasyTSF	0.5578	-1.1466	5.7628
ICMD	S3-Forecaster	0.1079	0.1628	0.5076
	S3-FastSketch	0.1377	-0.3635	0.4973
	ARIMA	19.3307	-2.689e+04	100.0000
	AR	0.3145	-6.1187	1.3408
	ETS	0.2181	-2.4226	0.8825
	Kernel Ridge	0.2650	-4.0541	1.2152
	Gaussian Process	0.2517	-3.5587	1.1824
	Chronos	0.1122	0.0939	0.5200
	LSTM	0.2552	-3.6854	1.1573
	CNN	0.2456	-3.3413	1.1250
	AR-KAN	0.1494	-0.6066	0.6820
	EasyTSF	0.2096	-2.1619	0.7993
Tourism	S3-Forecaster	0.0823	0.9636	0.8259
	S3-FastSketch	0.0616	0.9795	0.6308
	ARIMA	0.1585	0.8647	1.6908
	AR	0.1148	0.9290	1.1699
	ETS	0.0954	0.9509	0.9212
	Kernel Ridge	0.1478	0.8824	1.5375
	Gaussian Process	0.1153	0.9284	1.1658
	Chronos	0.1262	0.9143	1.3903
	LSTM	0.1409	0.8930	1.4096
	CNN	0.1652	0.8529	1.5454
	AR-KAN	0.0948	0.9516	0.8938
	EasyTSF	0.0687	0.9746	0.6857
M4	S3-Forecaster	0.1401	0.6866	1.3367
	S3-FastSketch	0.1122	0.7990	0.9240
	ARIMA	0.2521	-0.0144	2.5713
	AR	0.1918	0.4129	1.6391
	ETS	0.1427	0.6749	1.5010
	Kernel Ridge	0.1760	0.5055	1.6982
	Gaussian Process	0.1616	0.5832	1.6328
	Chronos	0.1554	0.6149	1.5043
	LSTM	0.1946	0.3956	1.9254
	CNN	0.1737	0.5184	1.6041
	AR-KAN	0.2560	-0.0455	2.4357
	EasyTSF	0.1943	0.3977	2.0007

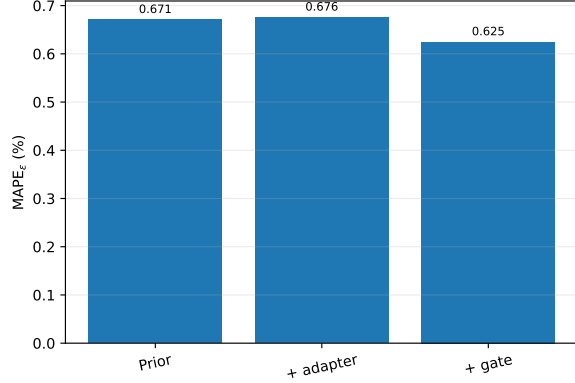


(a) ICMD

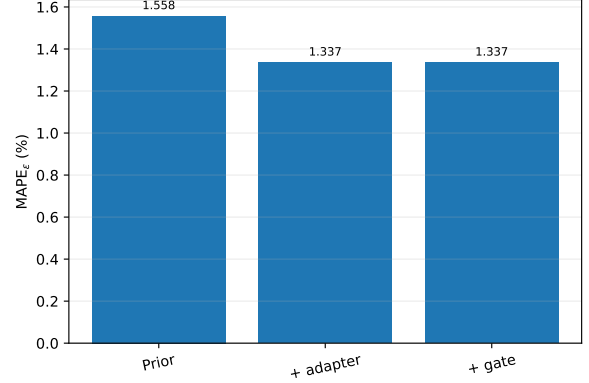


(b) M4

Figure 2: Primary test forecasts.



(a) ICMD



(b) M4

Figure 3: Component-wise ablation for the S3-Forecaster.

squared error is

$$\mathbb{E}[(e_t - g_t \hat{r}_t)^2 - e_t^2] = \mathbb{E}[g_t^2 \hat{r}_t^2] - 2\mathbb{E}[g_t e_t \hat{r}_t]. \quad (127)$$

A correction is useful only when its alignment with the future base residual is sufficiently large to offset the variance introduced by the correction itself. This condition is difficult to estimate reliably when $N < 200$, even when the readout is obtained through a stable convex regularized objective.

M4 provides the clearest adapter contribution: MAPE_e decreases from 1.5576 for the frozen prior to 1.3367 after residual adaptation, a relative reduction of approximately 14.2%. The gate remains equal to one, indicating that the improvement is driven by the residual representation and readout rather than by attenuation.

ICMD exhibits the complementary mechanism. The ungated adapter produces only a small MAPE degradation relative to the prior, but the OOB gate reduces MAPE from 0.6758 to 0.6246 and RMSE from 0.1465 to 0.1403, with a mean gate of 0.662. Hence, the gate provides its most substantive marginal contribution in the engineering case by suppressing over-correction. On M3, Tourism, and M4, the mean gate remains between 0.971 and 1.000 and therefore behaves approximately as an identity mapping.

CIF is an explicit negative-transfer case: the adapter increases error and the gate recovers only part of the loss. This result supports interpreting the gate as a protective shrinkage mechanism rather than a component that guarantees independent improvement.

Table 6: Component-wise ablation results for S3-Forecaster. Point metrics are unchanged by the uncertainty layer; ECP and MSIS are reported only for conformal variants.

Dataset	Variant	MAPE _{ϵ} ↓	RMSE ↓	Mean gate	ECP	MSIS ↓
CIF	Base only	0.3669	0.0310	–	–	–
	Base + residual adapter	0.5536	0.0460	1.000	–	–
	Base + residual + gate	<u>0.4449</u>	<u>0.0385</u>	0.870	–	–
	Full model + ACI	0.4449	0.0385	0.870	1.000	2.8417
	Full model + static conformal	0.4449	0.0385	0.870	0.750	2.0647
M3	Base only	3.4204	0.4324	–	–	–
	Base + residual adapter	3.7051	0.4716	1.000	–	–
	Base + residual + gate	<u>3.7042</u>	<u>0.4715</u>	0.998	–	–
	Full model + ACI	3.7042	0.4715	0.998	0.611	5.7616
	Full model + static conformal	3.7042	0.4715	0.998	0.611	5.7608
ICMD	Base only	<u>0.6708</u>	0.1593	–	–	–
	Base + residual adapter	0.6758	<u>0.1465</u>	1.000	–	–
	Base + residual + gate	0.6246	0.1403	0.662	–	–
	Full model + ACI	0.6246	0.1403	0.662	0.900	1.2908
	Full model + static conformal	0.6246	0.1403	0.662	0.900	1.2908
Tourism	Base only	0.7179	0.0681	–	–	–
	Base + residual adapter	0.8306	0.0828	1.000	–	–
	Base + residual + gate	<u>0.8259</u>	<u>0.0823</u>	0.971	–	–
	Full model + ACI	0.8259	0.0823	0.971	1.000	12.5348
	Full model + static conformal	0.8259	0.0823	0.971	1.000	17.6732
M4	Base only	1.5576	0.1509	–	–	–
	Base + residual adapter	1.3367	0.1401	1.000	–	–
	Base + residual + gate	1.3367	0.1401	1.000	–	–
	Full model + ACI	1.3367	0.1401	1.000	1.000	6.9809
	Full model + static conformal	1.3367	0.1401	1.000	0.944	4.8546

4.4. Adaptive Conformal Uncertainty Estimation

Table 7 combines adaptive conformal uncertainty metrics with operational runtime and trainable-parameter counts for the same dataset–model rows. In the uncertainty columns, ECP is interpreted through its absolute deviation from the nominal level 0.90, while MSIS remains the principal joint criterion because it penalizes both width and misses. Consequently, coverage above the nominal level is not intrinsically preferable when it is achieved through excessive width.

On M4, S3-Forecaster obtains the lowest MSIS (4.6380), but its ECP is 0.833. The result supports efficient adaptive conformal uncertainty estimation, not exact nominal coverage on a short test block. S3-FastSketch is second under MSIS on M4, which mirrors its strong point-forecasting performance but with slightly less favorable interval quality.

On ICMD, Chronos obtains the lowest MSIS (1.3580) with zero target-trainable parameters, while S3-Forecaster reaches ECP=1.0 but produces the largest MSIS, revealing over-conservative intervals on the model scale. Thus, the best point-forecasting behavior of S3-Forecaster on ICMD does not automatically transfer to interval efficiency.

M3 remains the clearest uncertainty limitation for S3-Forecaster: its ECP is 0.611, indicating that the calibration residuals are not representative of the subsequent test errors. LSTM and CNN obtain the strongest MSIS values on M3, whereas Chronos achieves high empirical coverage with wider intervals. CIF favors S3-Forecaster under MSIS, while Tourism favors S3-FastSketch and places Chronos behind ETS under the interval-score criterion. These results show that adaptive conformal estimation remains sensitive to both calibration representativeness and the point-forecast residual geometry.

Because the test blocks contain only 10–24 observations, empirical coverage is highly discrete. If k observations are covered in a test block of size T , and one additional miss changes ECP by $1/T$. The observed ECP values should therefore not be interpreted as exact finite-sample distribution-free predictive validity. The conformal construction avoids a parametric residual law, but its practical short-block validity depends on the sequential calibration conditions and on the evolution of the nonconformity scores.

Note that for ECP, bold and underlined entries denote the smallest and second-smallest absolute deviations from 0.90. For MSIS, runtime, and trainable parameters, lower is better. Interval width is reported descriptively. Time measures the operational fit-and-forecast stage after offline HPO; parameter counts exclude fixed reservoir matrices and frozen prior parameters. A one-observation change modifies ECP by 0.10 on ICMD, 0.083 on CIF, 0.056 on M3 and M4, and 0.042 on Tourism.

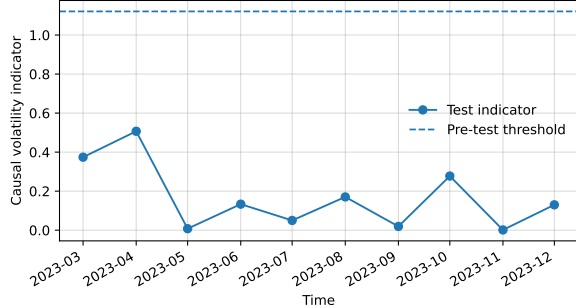
The causal volatility indicator was retained as a diagnostic of local distributional change. However, no test observation exceeded the fixed pre-test threshold in any dataset, as reported in Table 8. Overall metrics therefore coincide with the metrics of the regular test regime, and no separate extreme-regime performance estimate is available.

Table 7: Adaptive conformal uncertainty and operational-efficiency results on the model scale.

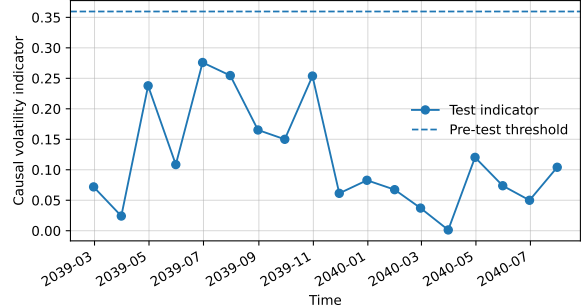
Dataset	Model	ECP $\rightarrow$ 0.90	IW	MSIS $\downarrow$	Time (s) $\downarrow$	Params $\downarrow$
CIF	S3-Forecaster	1.000	0.1098	1.4965	0.8168	554
	S3-FastSketch	1.000	0.1549	2.1119	1.7267	24
	ARIMA	1.000	0.1674	2.2810	0.3818	7
	AR	0.917	0.1658	2.2648	0.0078	2
	ETS	1.000	0.1959	2.6706	0.1134	18
	Kernel Ridge	1.000	0.2006	2.7341	0.0043	792
	Gaussian Process	1.000	0.2763	3.7665	0.5911	1,032
	Chronos	1.000	0.1524	2.0767	2.2445	0
	LSTM	0.833	0.2080	3.4810	4.6291	10,124
	CNN	1.000	0.4208	5.7350	5.1261	65,372
	AR-KAN	1.000	0.3996	5.4469	2.5985	199,975
	EasyTSF	1.000	0.1372	1.8696	1.5703	5
M3	S3-Forecaster	0.611	0.4658	4.3695	0.5483	1,082
	S3-FastSketch	0.944	1.5346	4.0887	3.7622	32
	ARIMA	0.944	2.3480	5.8136	0.1613	5
	AR	0.944	2.2877	5.2701	0.0109	3
	ETS	0.944	2.1689	4.2279	0.1692	18
	Kernel Ridge	0.944	2.2639	4.4071	0.0047	414
	Gaussian Process	1.000	1.9709	3.4225	0.6895	468
	Chronos	0.944	2.1647	4.1557	1.3550	0
	LSTM	1.000	1.8993	3.2980	16.0110	90,450
	CNN	1.000	1.9435	3.3747	4.0094	11,978
	AR-KAN	0.944	2.0599	3.6728	3.8750	79,116
	EasyTSF	0.944	2.1698	4.0885	0.3185	3
ICMD	S3-Forecaster	1.000	4.3752	11.1116	2.3646	802
	S3-FastSketch	0.900	0.4078	1.7512	9.5540	29
	ARIMA	0.900	38.9134	99.0639	0.0158	1
	AR	1.000	1.2365	3.1403	0.0084	15
	ETS	1.000	0.9636	2.4471	0.1062	18
	Kernel Ridge	1.000	0.9650	2.4507	0.0045	270
	Gaussian Process	0.900	0.7567	2.3255	0.5511	270
	Chronos	1.000	0.5347	1.3580	1.0396	0
	LSTM	1.000	0.8215	2.0863	9.9215	284,228
	CNN	1.000	0.8407	2.1351	3.7897	23,194
	AR-KAN	1.000	0.6425	1.6317	1.1184	18,840
	EasyTSF	0.800	0.5618	2.5675	0.1806	5
Tourism	S3-Forecaster	1.000	0.8064	10.1517	0.0761	842
	S3-FastSketch	0.792	0.1395	3.8261	4.1925	27
	ARIMA	0.958	0.6655	8.3783	1.0799	13
	AR	0.917	0.4346	6.0169	0.0127	28
	ETS	0.792	0.3014	4.7576	0.1461	18
	Kernel Ridge	0.750	0.3449	6.6261	0.0057	2,376
	Gaussian Process	1.000	0.8237	10.3698	4.6365	3,000
	Chronos	0.917	0.4287	5.6795	1.0096	0
	LSTM	0.875	0.4954	7.7034	11.7454	28,312
	CNN	0.917	0.6913	9.9082	12.3743	2,192
	AR-KAN	1.000	0.6234	7.8473	2.8422	219,483
	EasyTSF	1.000	0.4164	5.2420	1.5747	20
M4	S3-Forecaster	0.833	0.4227	4.6380	0.0823	182
	S3-FastSketch	0.778	0.2964	4.7737	0.9148	27
	ARIMA	1.000	1.1254	9.2515	1.9694	10
	AR	1.000	1.7706	14.5559	0.0337	47
	ETS	1.000	1.3009	10.6950	0.1534	18
	Kernel Ridge	1.000	1.0498	8.6304	0.1398	7,542
	Gaussian Process	1.000	1.2014	9.8763	32.7088	7,848
	Chronos	1.000	1.1137	9.1554	1.4109	0
	LSTM	1.000	1.1817	9.7147	12.6594	6,002
	CNN	1.000	1.1869	9.7574	9.0094	42,306
	AR-KAN	1.000	1.4715	12.0972	6.0473	3,312
	EasyTSF	1.000	1.4474	11.8990	0.9907	25

Table 8: Causal volatility diagnostics.

Dataset	Test size	Threshold τ	Maximum indicator	Exceedances
CIF	12	0.1624	0.0953	0
M3	18	1.5929	1.1934	0
ICMD	10	1.1210	0.5069	0
Tourism	24	0.2398	0.1851	0
M4	18	0.3596	0.2760	0



(a) ICMD



(b) M4

Figure 4: Causal volatility diagnostics.

4.5. Data Efficiency and History-Length Spectrum

The reduced-history experiments identify a dataset-dependent operating regime. The panels report MAPE in percentage units as a function of the available history length, so they should be read as trajectory-level evidence rather than as a duplicate of the single split in Table 5.

Across ICMD and M4, the S3 family occupies the most favorable region of the curves over substantial portions of the history spectrum. This is consistent with the benchmark table: S3-Forecaster is strongest on ICMD for variance-sensitive point accuracy, whereas S3-FastSketch is strongest on M4 and also competitive in ICMD proportional error. The profiles also show that adding observations is not automatically beneficial; gains appear when the additional history is relevant to the current regime and when the residual representation has enough support to estimate a stable readout.

Tourism exhibits a crossover pattern rather than a single dominant curve. S3-FastSketch, S3-Forecaster, EasyTSF, and ETS exchange relative positions as history length changes, which is compatible with an approximation–estimation trade-off. Compact residual features can control variance at shorter or intermediate histories, while more conventional or trained alternatives can become competitive once their parameters are sufficiently identified.

CIF and M3 again expose limitations. CIF shows that the S3 family can be strong on the final split while still exhibiting sensitivity to which historical window is available. M3 spans a narrow history range and does not provide enough variation to infer a stable data-efficiency crossover. A useful conceptual quantity is

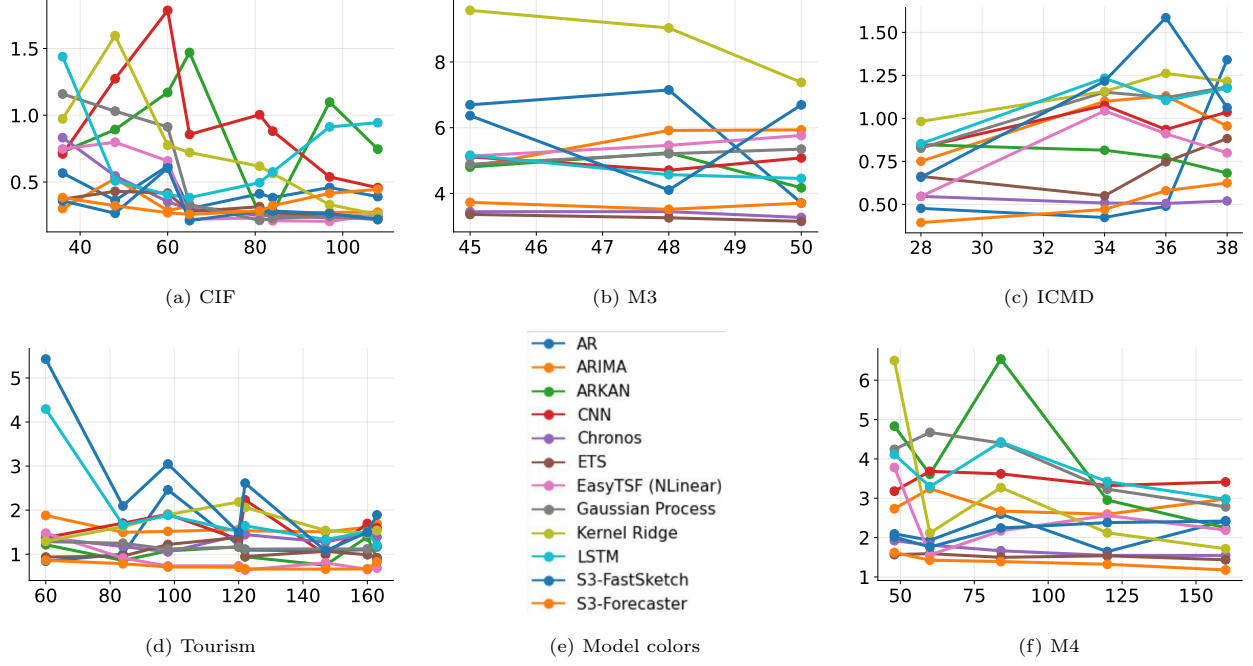


Figure 5: Data-efficiency profiles. The horizontal axis shows the available history and the vertical axis reports MAPE as a percentage.

therefore the effective history size

$$N_{\text{eff}} = \frac{\left(\sum_{t=1}^N w_t\right)^2}{\sum_{t=1}^N w_t^2}, \quad (128)$$

where w_t denotes the relevance of observation t to the current regime. The results indicate that S3-family models benefit from additional data only when the added observations increase N_{eff} , not merely nominal N .

4.6. Multi-Prior Robustness and Backbone Transferability

The prior-transfer experiment evaluates Rolling, ETS, and Prophet backbones. The multi-prior panels reinforce architectural compatibility, but they also make clear that performance is not backbone invariant. Across datasets, residual adaptation is useful only when the chosen prior leaves a stable and causally predictable residual component. When the prior already captures the main dynamics, or when its residuals change regime across train, calibration, and test, the adapter can add variance rather than reduce error.

The qualitative pattern is consistent with the benchmark and data-efficiency figures. ICMD and M4 provide the strongest evidence that an S3 correction can be useful under favorable residual structure, but CIF, M3, and Tourism show that changing the backbone can reverse the ordering among S3-Forecaster, S3-FastSketch, and the prior-only alternatives. The model-color panel in Figure 6 should therefore be read as a map of backbone sensitivity rather than as evidence for a universally transferable adapter.

Let the prior residual admit the decomposition

$$e_t = m(\mathbf{h}_{t-1}) + \xi_t, \quad (129)$$

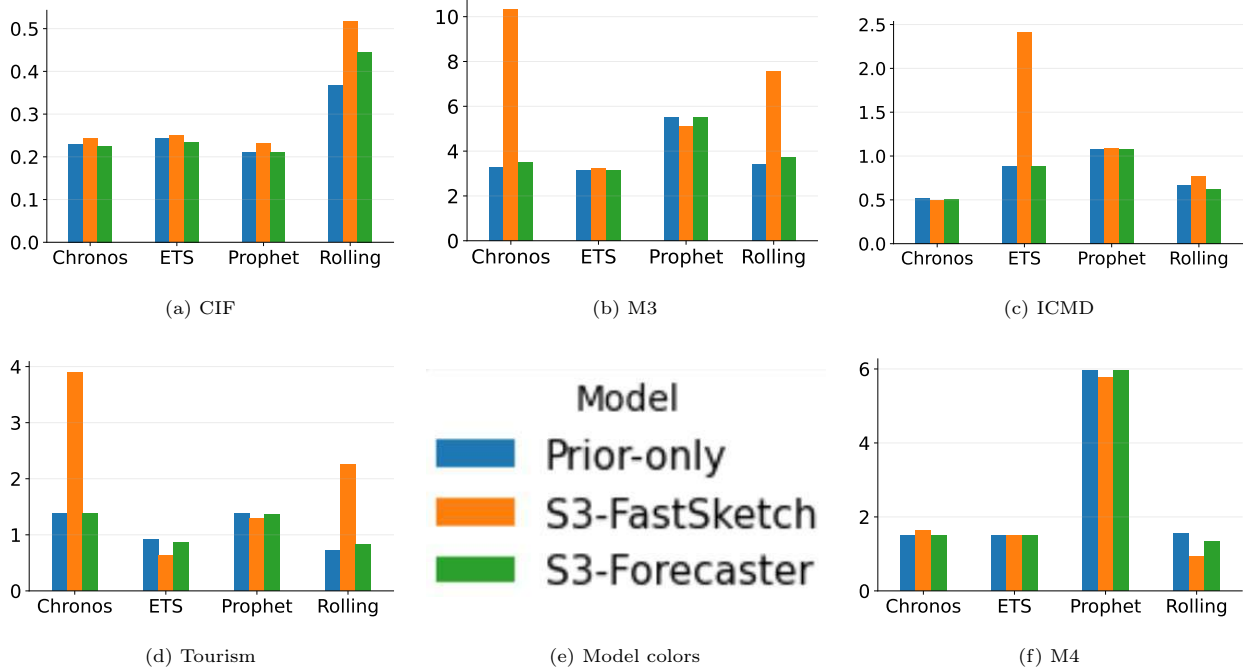


Figure 6: Multi-prior transferability. The vertical axis reports MAPE as a percentage for each model–prior configuration.

where $m(\mathbf{h}_{t-1})$ is the component predictable from the causal state and ξ_t is an innovation. The potential value of residual adaptation is governed by

$$\text{Var} [\mathbb{E} (e_t \mid \mathbf{h}_{t-1})] . \quad (130)$$

When this quantity is small, the adapter mainly adds estimation variance. When it is large and stable across train, calibration, and test, the adapter can correct systematic prior mismatch.

A severe mismatch exceeds the ESN’s corrective capacity when the residual trajectory lies outside the calibration support, changes sign across regimes, or contains temporal structure absent from the fixed reservoir state space. Neither the gate nor the convex readout can reconstruct information that is not represented in the causal features. This explains why architectural transferability should not be interpreted as backbone-independent accuracy.

Although Chronos is included in the direct benchmark, the prior-transfer experiment does not include Chronos as a backbone within the S3 adaptation pipeline. Accordingly, the prior-transfer results should not be interpreted as a direct test of foundation-model residual mismatch.

4.7. Operational Efficiency and Parameter Economy

The operational-efficiency columns in Table 7 report runtime after the stored dataset-level configurations have been loaded. On the model-scale benchmark, S3-Forecaster has a median runtime of approximately 0.55 s and a median of 802 trainable coefficients. Across datasets with successful paired executions, the median runtime ratios relative to S3-Forecaster are approximately $21\times$ for LSTM, $9\times$ for CNN, and $5\times$

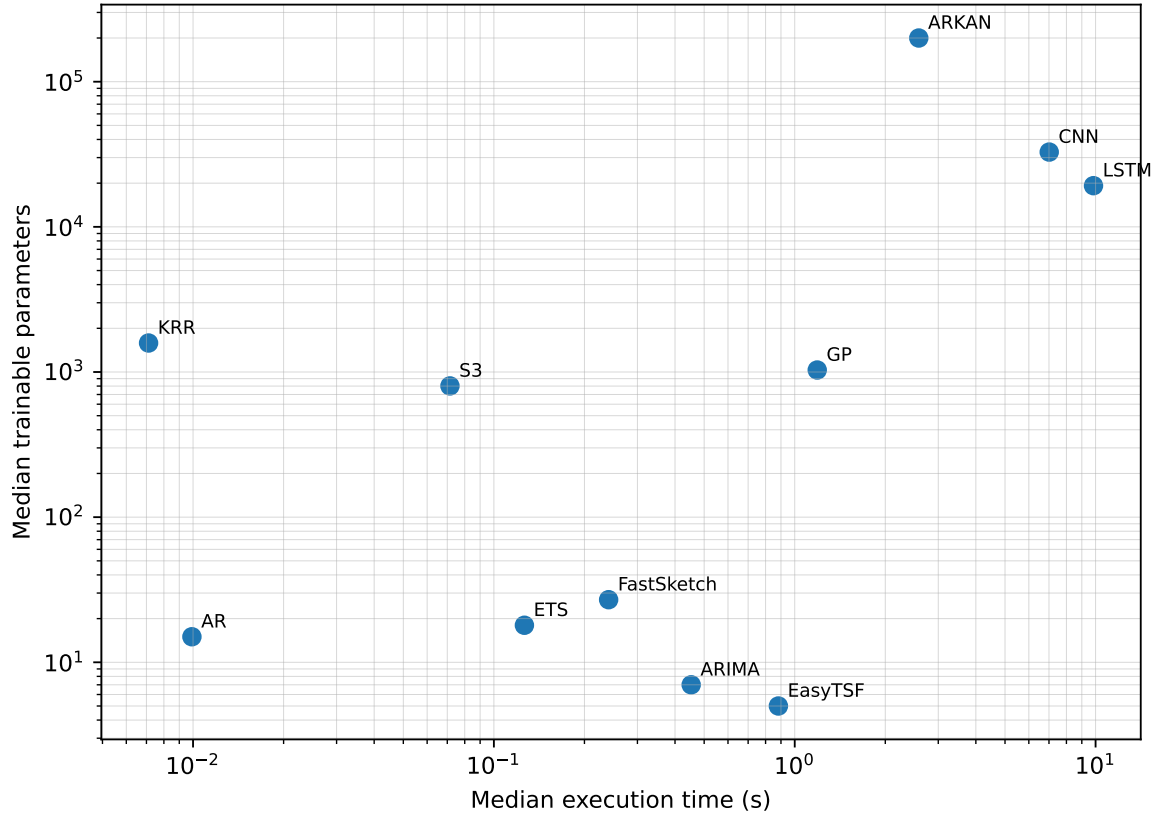


Figure 7: Operational runtime-parameter trade-off.

for AR-KAN. Chronos has zero target-trainable coefficients in this accounting, but its median operational runtime is still about $2.5\times$ that of S3-Forecaster.

S3-Forecaster is not the smallest or fastest model in absolute terms. Chronos, AR, Kernel Ridge, ETS, and EasyTSF can require fewer learned coefficients or lower latency in specific settings. Its engineering advantage is instead the joint profile of nonlinear fixed-state representation, moderate target-domain complexity, convex readout estimation, and the absence of iterative backpropagation through the feature extractor. Because fixed reservoir matrices and frozen prior parameters are excluded from the trainable-parameter count, the parameter column measures statistical adaptation complexity rather than total memory footprint.

4.8. Limitations and Scientific Synthesis

The complete evidence supports a selective rather than universal interpretation of the S3 family. ICMD remains the principal engineering case: S3-Forecaster achieves the strongest model-scale RMSE and R^2 , S3-FastSketch gives the best proportional point error, and Chronos gives the best interval score. This split shows that point accuracy, proportional error, and interval efficiency can favor different mechanisms even on the same industrial series.

M4 provides the principal reproducible public result for the S3 family: S3-FastSketch is strongest in point forecasting, while S3-Forecaster gives the best adaptive conformal MSIS. CIF supports the value of the larger

reservoir, whereas Tourism favors the compact sketch representation. M3 remains a boundary case in which S3-Forecaster ties ETS in point metrics but under-covers in adaptive conformal evaluation.

5. Conclusions

This study introduced S3-Forecaster and S3-FastSketch, two gradient-free residual-adaptation architectures for engineering time series with fewer than 200 observations. Both methods preserve a frozen causal forecasting prior and learn only a regularized correction of its residual errors. S3-Forecaster represents the residual dynamics through a fixed Echo State Network, a convex Bayesian Ridge readout, and a volatility-sensitive out-of-bag gate, whereas S3-FastSketch replaces the reservoir with a deterministic causal multiscale representation and projected calibration shrinkage. Adaptive Conformal Inference provides a common distribution-free uncertainty layer without assuming a parametric residual distribution or requiring end-to-end neural fine-tuning.

The empirical evidence supports dataset- and objective-dependent model selection rather than universal superiority. In the ICMD engineering application, S3-Forecaster achieved the lowest model-scale RMSE and the highest R^2 , while S3-FastSketch produced the lowest proportional point error and Chronos obtained the most favorable MSIS. On M4, S3-FastSketch was the strongest point forecaster across RMSE, R^2 , and MAPE_ε , whereas S3-Forecaster yielded the best adaptive conformal interval score. CIF favored the reservoir-based variant, Tourism favored the compact sketch representation, and M3 constituted a boundary case in which S3-Forecaster matched ETS in point accuracy but exhibited insufficient adaptive conformal coverage. Consequently, the appropriate S3 variant depends on the residual geometry, the evaluation criterion, and the stability of the calibration regime.

The ablation results clarify the mechanism underlying these outcomes. Residual adaptation is beneficial only when the frozen prior leaves a stable component that is causally predictable from the available residual history. When this condition is not satisfied, the correction may introduce more estimation variance than systematic-error reduction. The out-of-bag gate should therefore be interpreted as protective shrinkage rather than as an independent source of accuracy: it substantially attenuated over-correction in ICMD, behaved approximately as the identity in several favorable regimes, and only partially mitigated negative transfer in CIF. Similarly, the effectiveness of ACI depended on the representativeness of the short calibration block; its sequential update improved interval adaptation in selected cases but did not guarantee exact conditional coverage over test blocks containing only 10–24 observations.

From an operational perspective, the proposed family achieves low target-domain statistical complexity by replacing iterative backpropagation with fixed feature transformations and closed-form convex readouts. S3-Forecaster required a median operational runtime of approximately 0.55 s and was, in median paired comparisons, approximately $21\times$, $9\times$, and $5\times$ faster than the evaluated LSTM, CNN, and AR-KAN models, respectively. S3-FastSketch reduced the learned adaptation to approximately 24–32 coefficients in the reported experiments, although its wall-clock advantage was not uniform across datasets. Thus, the principal

engineering contribution is not absolute minimal latency, but a favorable balance among nonlinear residual representation, controlled trainable complexity, causal uncertainty adaptation, and deployment without neural-network retraining.

Overall, the results position the S3 family as a selective and computationally efficient residual-adaptation framework for short, non-stationary engineering time series. The current evidence remains limited by the small number of evaluated series, discrete empirical coverage estimates, one-step-ahead forecasting, and sensitivity to prior and calibration selection. Future work should therefore evaluate the framework over complete multi-series collections, extend the architecture to multivariate and multi-horizon forecasting, incorporate exogenous engineering variables, and develop diagnostics that estimate residual predictability before activating the adapter. Such extensions would enable automatic selection among the frozen prior, S3-Forecaster, and S3-FastSketch according to the effective information content of the available history.

CRedit authorship contribution statement

Alejandro Patino Bedoya: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Project administration, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The M3, M4, Tourism, and CIF 2016 benchmark datasets used in this study are publicly available from the original sources cited in the manuscript. The source code, selected public-series identifiers, truncation lengths, random seeds, serialized hyperparameter configurations, and scripts required to reproduce the public benchmark, ablation, data-efficiency, and multi-prior experiments are openly available at <https://github.com/AlejoPatigno/S3Forecaster-S3FastSketch>.

The Industrial Construction-Materials Demand dataset cannot be made publicly available because the underlying transactional records contain confidential commercial information and are subject to data-use restrictions imposed by the data owner. The manuscript reports the target definition, monthly aggregation procedure, chronological partitions, preprocessing protocol, model configurations, and evaluation procedure required to reproduce the methodological workflow. Access to the proprietary records cannot be guaranteed and would require explicit authorization from the data owner.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI’s ChatGPT to support literature organization, language refinement, and structural editing. After using this tool, the author reviewed, verified, and edited the content as needed and takes full responsibility for the content of the published article.

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